Project: Enterprise Perimeter Security Architecture

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Environment: Palo Alto Networks VM-Series (EDU-210 Lab 17 Capstone | NDG NetLab+)

Focus: Layer 3 Segmentation, NAT, Security Policies, and Threat Inspection Profiles

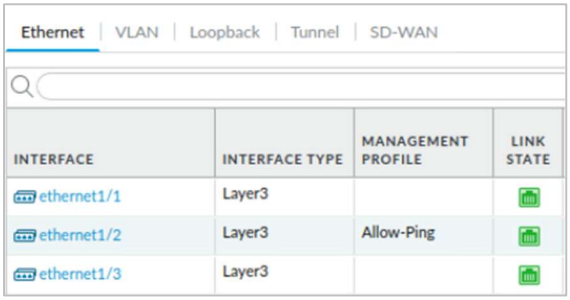
Palo Alto Networks EDU-210 Lab 17 Capstone: Enterprise Perimeter Security Architecture

This documentation outlines my implementation workflow and technical notes for the Palo Alto Networks EDU-210 Lab 17 Capstone, completed within the NDG NetLab+ environment. Rather than building an entire topology completely from scratch, this project focused on inheriting a pre-configured baseline and expanding it into a fully secured enterprise perimeter. Throughout the lab, I established proper Layer 3 network segmentation, configured virtual routing and source NAT translations, enforced application-level security policies, and implemented a comprehensive suite of threat-inspection profiles to lock down the perimeter.

Load Lab Configuration, Interfaces Created, & Configured (2 photos)

To get started, I accessed the firewall management web interface and navigated to Device, Setup, and Operations. By loading the predefined configuration snapshot named edu-210-lab-17a-start.xml and committing the changes, I initialized the baseline environment required for the capstone objectives.

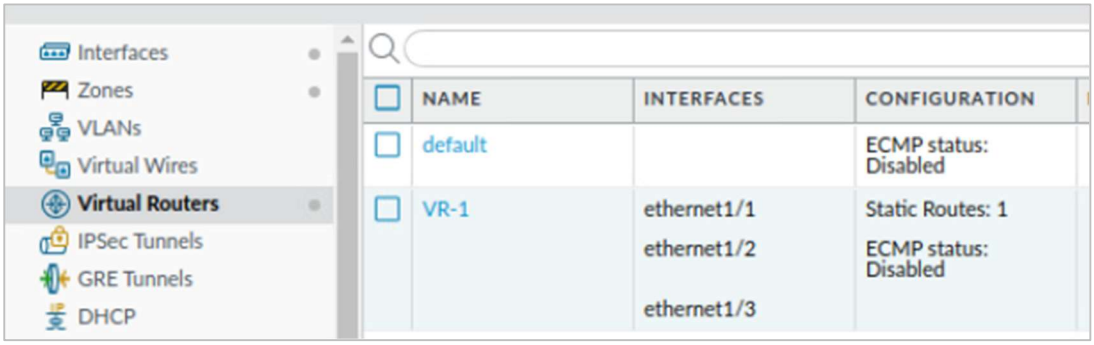
Proper network segmentation starts at the physical interface layer. I converted three firewall ports into Layer 3 interfaces to separate traffic into distinct security zones. I assigned ethernet1/1 the IP address 203.0.113.20/24 to serve as the external internet WAN connection. For ethernet1/2, which manages the internal user network, I configured 192.168.1.1/24 and attached an Allow-ping management profile to allow for network diagnostics. Finally, I set ethernet1/3 to 192.168.50.1/24 to support an isolated extranet partner server segment.



IP ADDRESS	VIRTUAL ROUTER	TAG	VLAN / VIRTUAL-WIRE	SECURITY ZONE
203.0.113.20/24	VR-1	Untagged	none	Internet
192.168.1.1/24	VR-1	Untagged	none	Users
192.168.50.1/24	VR-1	Untagged	none	Extranet

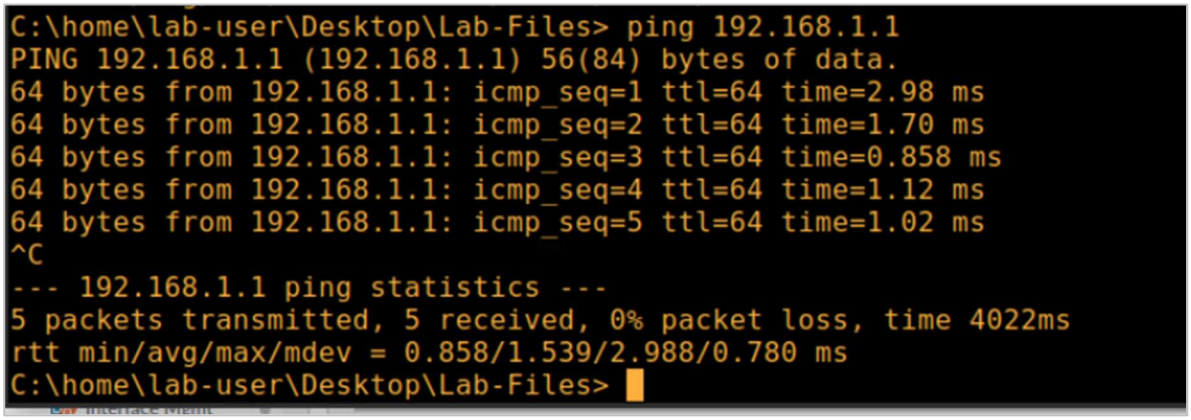
Virtual Router & Static Route Created & Configured

Once the interfaces were up, they needed a mechanism to share routing tables and direct traffic. I created a virtual router named VR-1 and bound all three Layer 3 interfaces to it. To ensure outbound traffic could actually exit the local network, I added a static default route named Default-Route pointing toward the internet gateway.



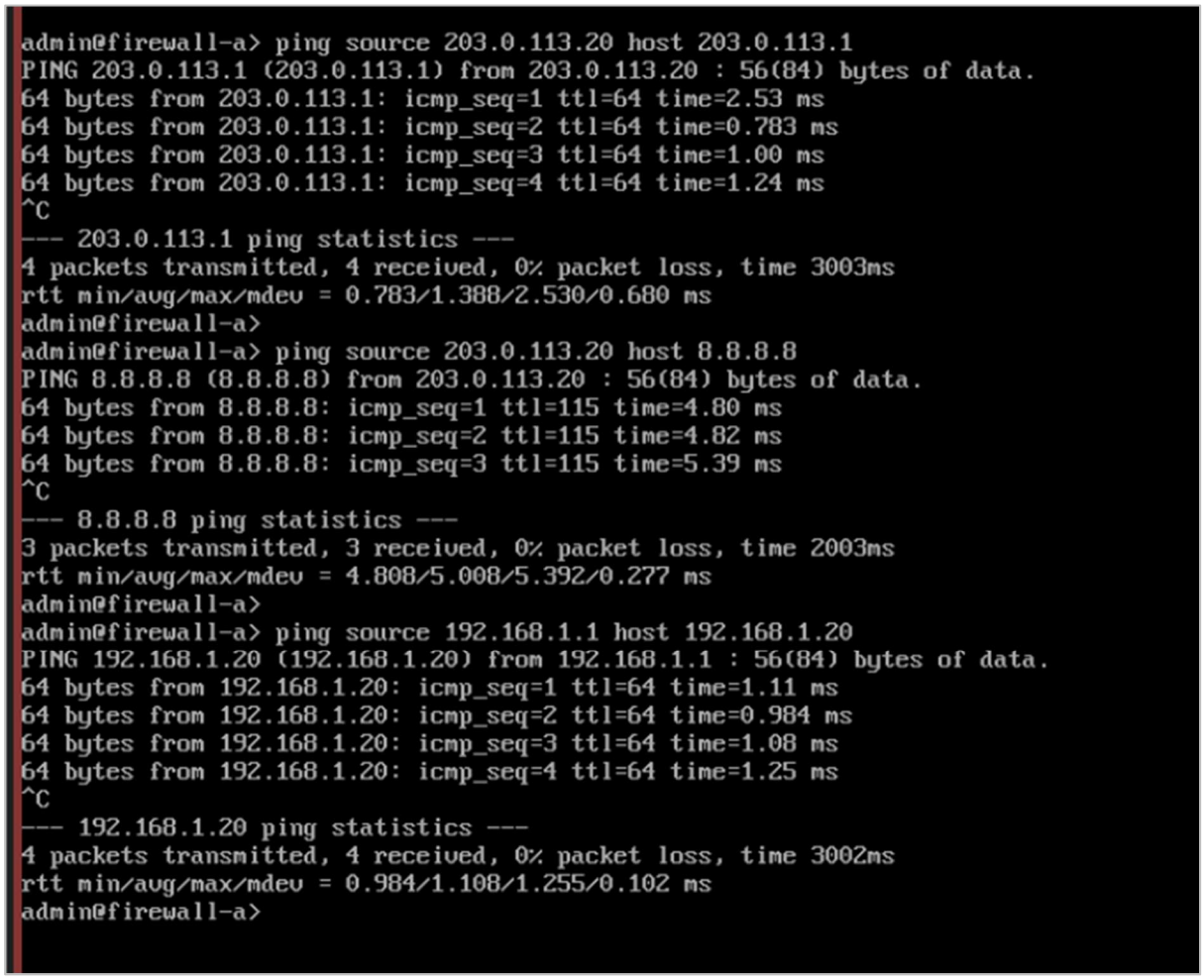
Internal Host Ping to 192.168.1.1 Confirmed Working

Before moving forward, I wanted to verify that internal routing was functioning correctly at the local level. Opening the command line from the internal client host, I pinged the firewall gateway interface at 192.168.1.1. The test returned zero percent packet loss with minimal latency, confirming that the client could successfully reach its gateway across the Layer 3 boundary.



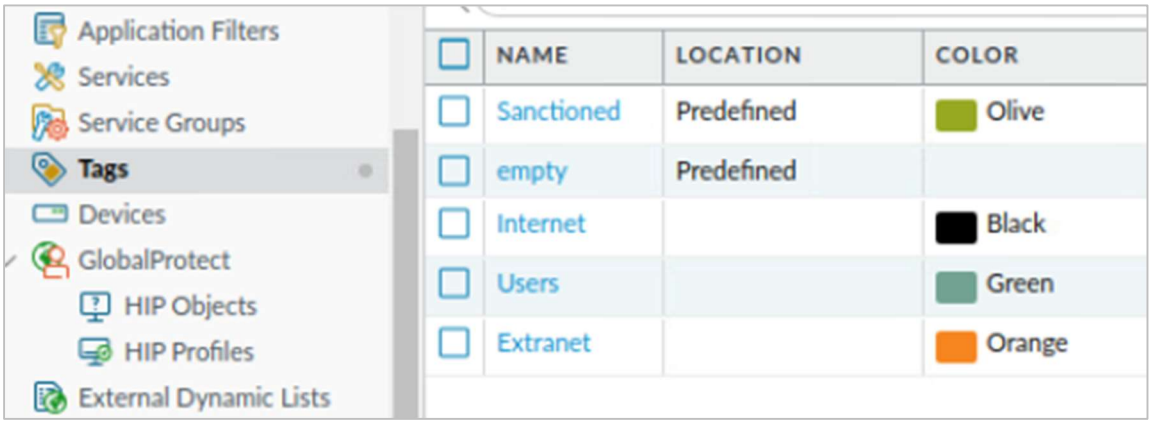
Ping Sources Confirmed Working

To validate routing paths across the entire firewall architecture, I logged into the firewall command line interface to execute source-specific pings. I tested from the external interface IP 203.0.113.20 to its gateway and out to public DNS at 8.8.8.8, as well as from the internal interface 192.168.1.1 to the client host at 192.168.1.20. All test cases returned zero percent packet loss, verifying that the source bindings, routing tables, and network paths were operating properly.



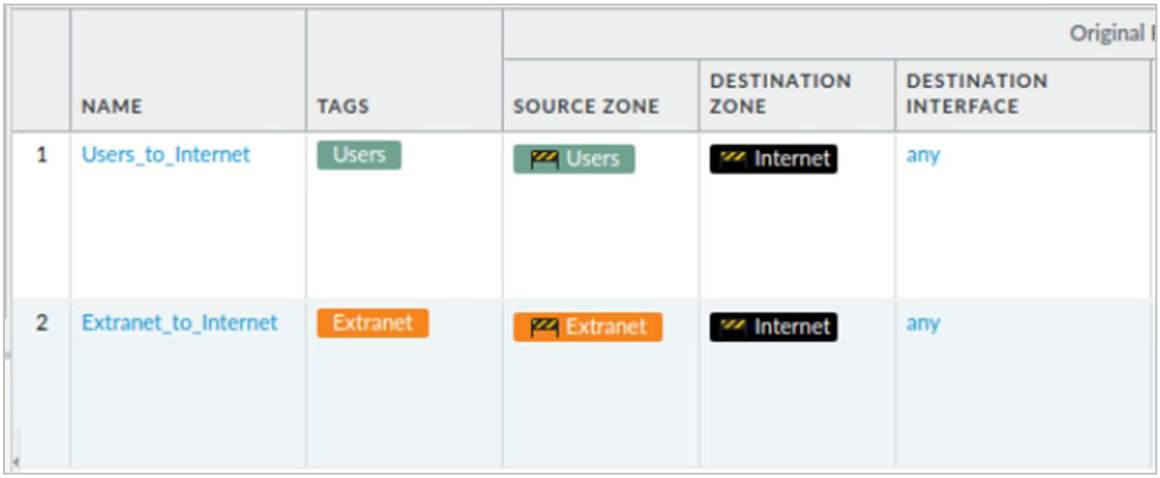
Tags Created

Maintaining visibility across a busy firewall dashboard is crucial in production environments. I created custom administrative tags to clearly categorize my security zones, assigning black to the Internet zone, green to the Users zone, and orange to the Extranet zone.



NAT Policies Created & Configured (2 Photos)

Because internal subnets rely on private IP addressing, source NAT rules were necessary to allow outbound internet connectivity without exposing internal topology. I created two source NAT rules: Users_to_Internet and Extranet_to_Internet. Both rules utilize dynamic IP and port translation (PAT) mapped to the external WAN interface ethernet1/1 using the public IP address 203.0.113.20, allowing internal hosts to safely access external resources while conserving public IP space.



DESTINATION INTERFACE	SOURCE ADDRESS	DESTINATION ADDRESS	SERVICE	SOURCE TRANSLATION
any	any	any	any	dynamic-ip-and-port ethernet1/1 203.0.113.20/24
any	any	any	any	dynamic-ip-and-port ethernet1/1 203.0.113.20/24

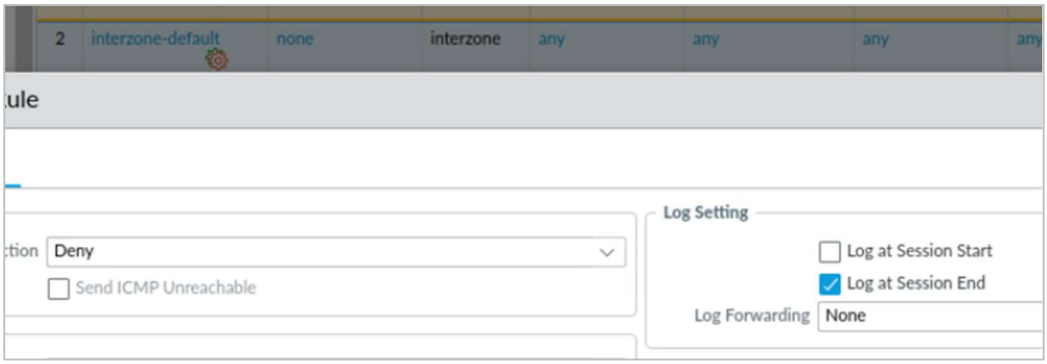
Allow & Block Tags Created

To improve rulebase readability and make policy auditing more intuitive, I added two color-coded visual markers: an Allow tag set to lime green and a Block tag set to red.



Interzone-default Overridden & Configured Properly

Palo Alto firewalls deny interzone traffic by default out of the box. To ensure proper auditing and visibility for unauthorized traffic crossing between zones, I modified the built-in interzone-default rule to enable session-end logging (Log at Session End), ensuring any unclassified connection attempts are explicitly recorded in the system logs for security monitoring.

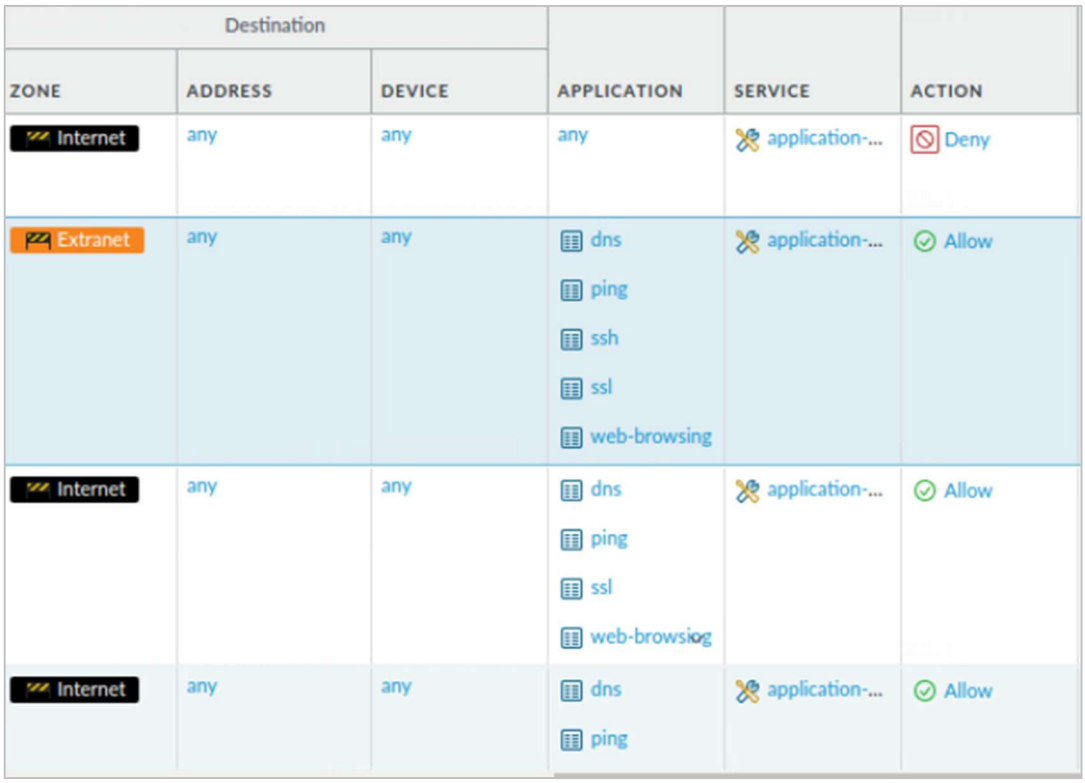


Security Policies Configured (2 photos)

Moving away from legacy, insecure port-based rules, I built granular security policies leveraging Palo Alto's application-filtering engine. My first policy, Block_Bad_URLs, explicitly denies outbound traffic matching high-risk categories including hacking, phishing, malware, and unknown threats. The remaining rules—Users_to_Extranet, Users_to_Internet, and Extranet_to_Internet—permit precisely defined application protocols like dns, ping, ssh, ssl, and web-browsing between authorized zones.



	NAME	TAGS	TYPE	Source			
				ZONE	ADDRESS	USER	DEVICE
1	Block_Bad_URLs	Block	universal	Extranet Users	any	any	any
2	Users_to_Extranet	Allow	universal	Users	any	any	any
3	Users_to_Internet	Allow	universal	Users	any	any	any
4	Extranet_to_Internet	Allow	universal	Extranet	any	any	any



Destination			APPLICATION	SERVICE	ACTION
ZONE	ADDRESS	DEVICE			
Internet	any	any	any	application-...	Deny
Extranet	any	any	dns ping ssh ssl web-browsing	application-...	Allow
Internet	any	any	dns ping ssl web-browsing	application-...	Allow
Internet	any	any	dns ping	application-...	Allow

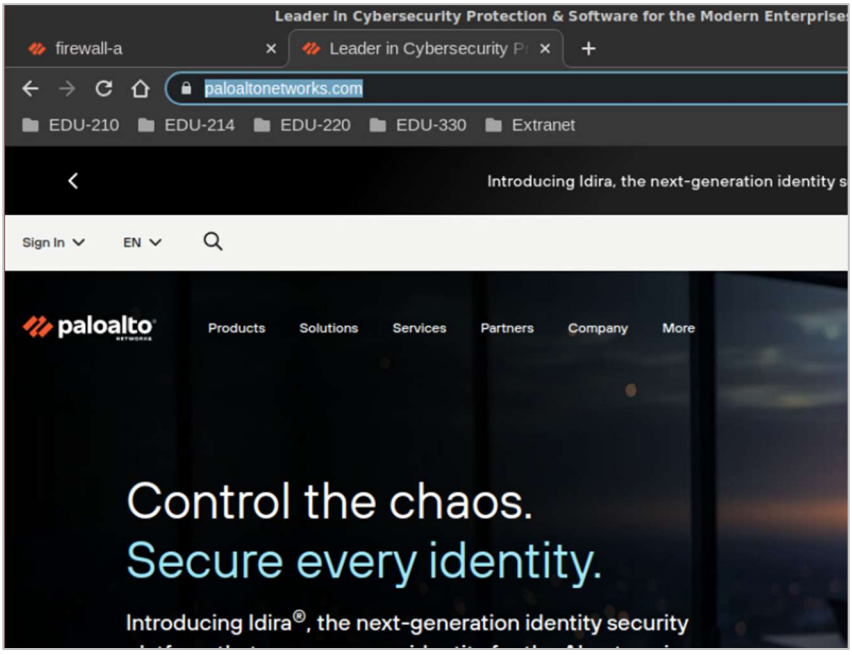
Ping Googles DNS & google.com Confirmed

With the security and NAT policies in place, I tested outbound connectivity from the client host command line by pinging Google's public DNS (8.8.8.8) and querying google.com. Both tests completed with zero percent packet loss, validating that DNS resolution and external routing rules were functioning as intended.

```
C:\home\lab-user\Desktop\Lab-Files> ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=114 time=5.87 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=114 time=4.68 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=114 time=4.68 ms
^C
--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 4.685/5.082/5.876/0.566 ms
C:\home\lab-user\Desktop\Lab-Files>
C:\home\lab-user\Desktop\Lab-Files> ping google.com
PING google.com (172.253.115.113) 56(84) bytes of data.
64 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=1 ttl=106 time=4.96 ms
64 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=2 ttl=106 time=5.33 ms
64 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=3 ttl=106 time=5.04 ms
^C
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 4.964/5.114/5.336/0.170 ms
C:\home\lab-user\Desktop\Lab-Files>
```

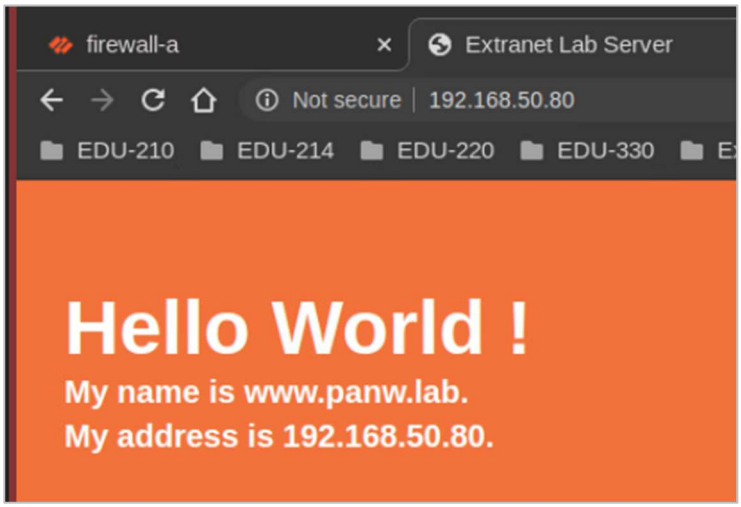
Able to Access https://paloaltonetworks.com

To verify secure web traffic handling, I opened a browser on the client host and navigated to paloaltonetworks.com. The site loaded successfully over valid SSL encryption, confirming that the web-browsing security policy and NAT rules were operating correctly.



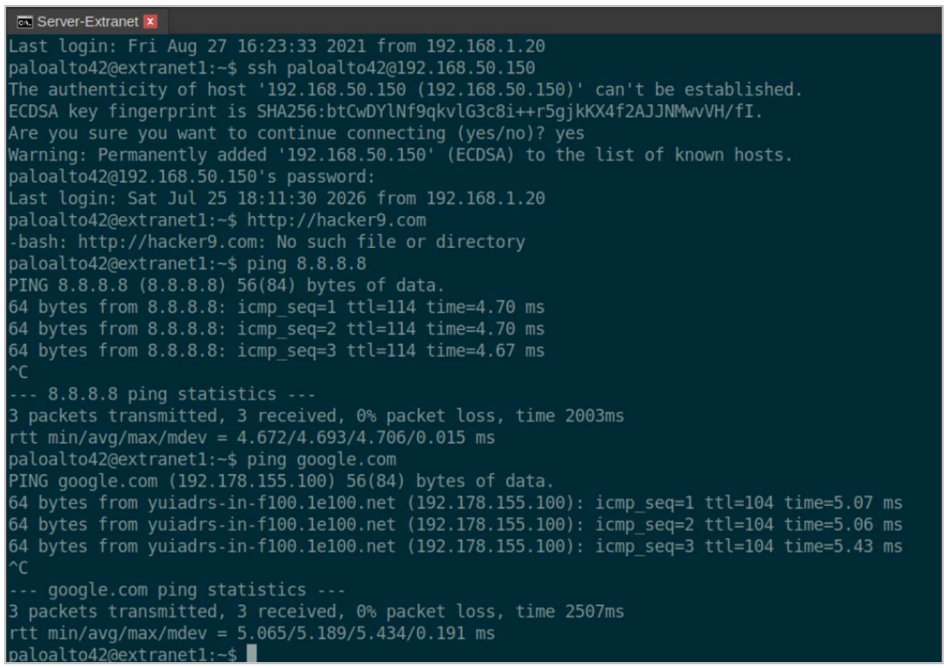
Able to Connect http://192.168.50.80

I also tested internal access to the isolated extranet environment by browsing to the web server at 192.168.50.80. The browser successfully displayed the welcome page, verifying that the Users_to_Extranet policy correctly permitted traffic across the zone boundary.



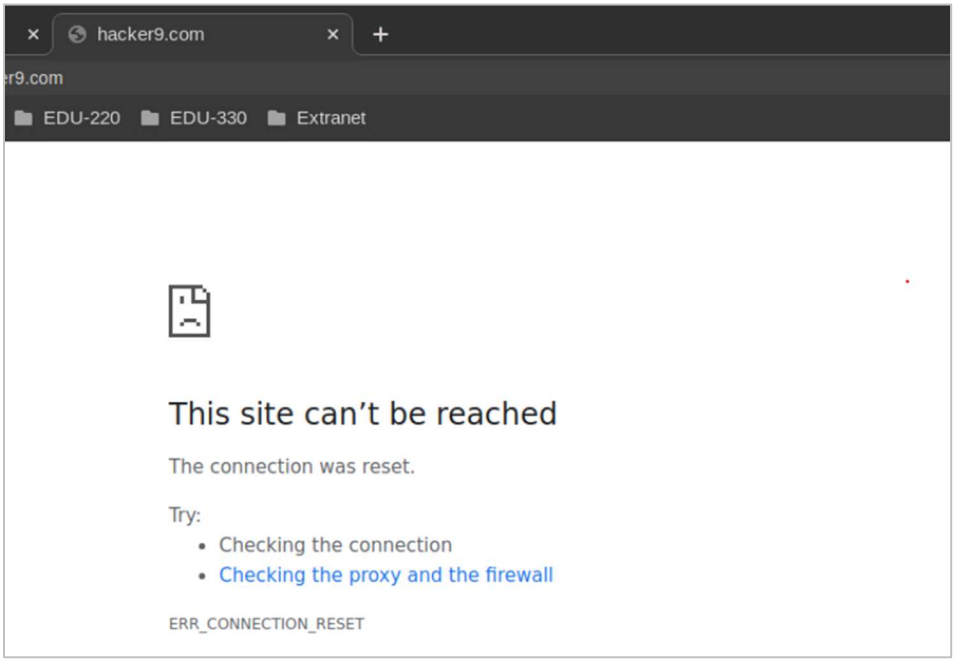
Connected to SSH, pinged 8.8.8.8 & google.com

To test terminal-level access and extranet-zone routing, I connected to the extranet host at 192.168.50.150 via SSH using the lab-provided account credentials. Once authenticated successfully, I ran pings to 8.8.8.8 and google.com from the server terminal; both tests succeeded with zero percent packet loss, confirming that outbound routing from the extranet segment was fully operational.



Hacker9.com Blocked

To validate negative security constraints and URL filtering logic, I attempted to browse to hacker9.com from the internal client. The connection was immediately reset and blocked, demonstrating that the firewall successfully prevents users from accessing unauthorized or malicious sites categorized under hacking and malware.



URL Filtering Profile Created & Set to Alerts

To begin implementing deep packet inspection, I created a corporate URL filtering profile named Corp-URL using the default profile as a baseline. I configured it to log access across web categories, allowing normal web traffic to pass while ensuring the firewall actively captures and monitors browsing behavior.



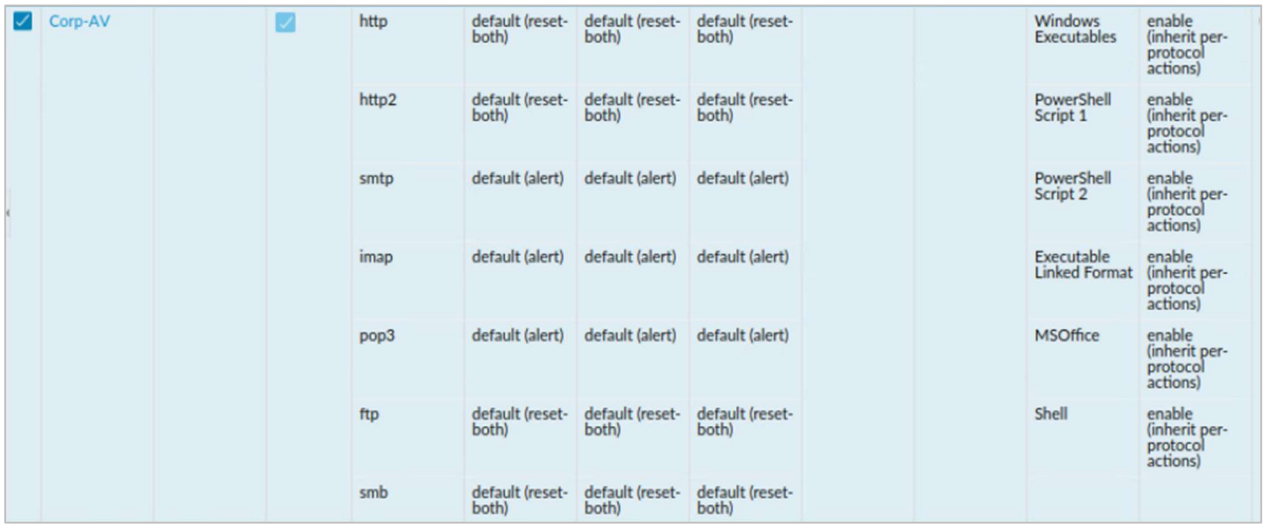
Corp-FB Created & Set to Block All Risky File Types

For payload and file control, I established a file-blocking profile called Corp-FB using a strict template. This profile is configured to block risky file types and encrypted archives while logging other file transfers, preventing users from downloading hazardous executable payloads or hidden files that could bypass standard inspection.



Corp AV Created as Antivirus Security Profile

Next, I configured an antivirus profile named Corp-AV built on the default profile baseline. I set core protocols like HTTP, HTTP2, FTP, and SMB to trigger a reset-both action upon virus detection, configured email protocols to generate alerts, and enabled inspection for high-risk scripts, Windows executables, ELF files, and Office documents.



Corp-AS Created as Anti-Spyware Security Profile

To defend against telemetry and callbacks, I built an anti-spyware profile named Corp-AS using a strict template. For critical, high, and medium severity threats, I configured the action to reset-both so the firewall actively terminates malicious connections and command-and-control communication

☐	Corp-AS		Policies: 5	simple-critical	any	critical	reset-both	disable
				simple-high	any	high	reset-both	disable
				simple-medium	any	medium	reset-both	disable
				simple-informational	any	informational	default	disable
				simple-low	any	low	default	disable

Corp-Vuln Created as Vulnerability Protection Security Profile

I implemented a vulnerability protection profile named Corp-Vuln using a strict template comprising ten inspection rules. For critical, high, and medium threats targeting clients or servers, the action is set to reset-both, ensuring known exploit signatures are dropped and compromised sessions are torn down immediately.

☑	Corp-Vuln		Rules: 10	simple-client-critical	any	client	critical	reset-both	disable
				simple-client-high	any	client	high	reset-both	disable
				simple-client-medium	any	client	medium	reset-both	disable
				simple-client-informational	any	client	informational	default	disable
				simple-client-low	any	client	low	default	disable
				simple-server-critical	any	server	critical	reset-both	disable
				simple-server-high	any	server	high	reset-both	disable
				more...					

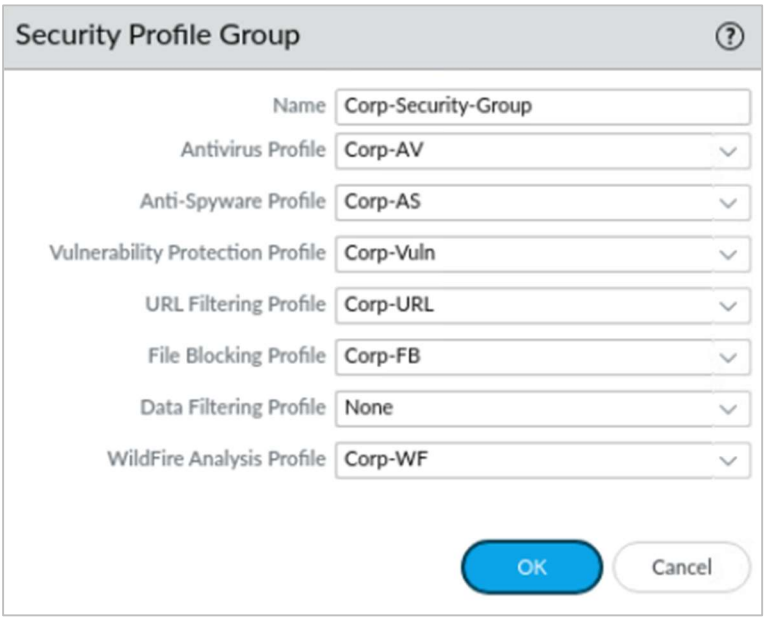
Corp-WF Created as WildFire Profile

To protect against zero-day threats and unknown binaries, I created a WildFire analysis profile named Corp-WF utilizing the default profile as a base. This profile enables automated forwarding of unknown files to the Palo Alto WildFire cloud sandbox for dynamic behavioral analysis.

	NAME	ANALYSIS	LOCATION	RULE NAME	APPLICATIONS
☐	default	public-cloud	Predefined	default	any
☑	Corp-WF	public-cloud		default	any

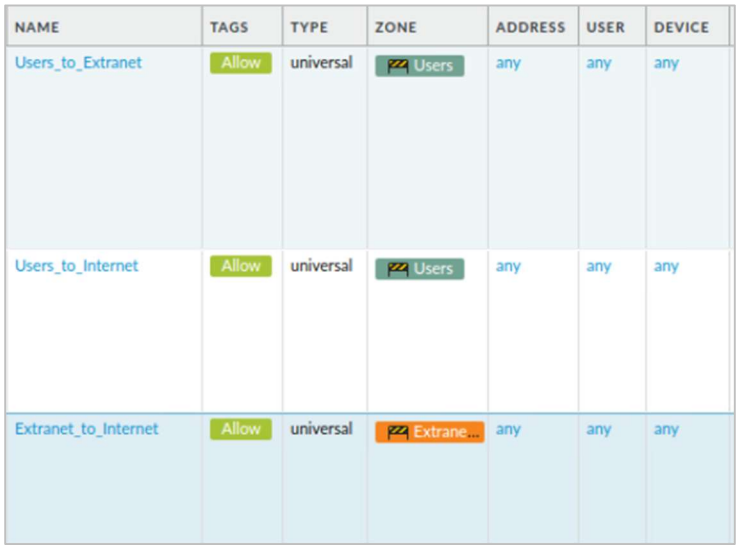
Corp-Security-Group Created & Configured

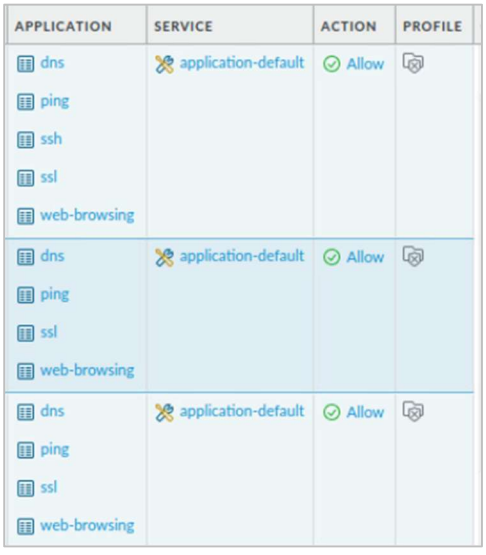
Rather than applying individual security profiles manually to every policy rule, I bundled the individual inspection profiles (Corp-AV, Corp-AS, Corp-Vuln, Corp-URL, Corp-FB, and Corp-WF) into a single profile group named Corp-Security-Group.

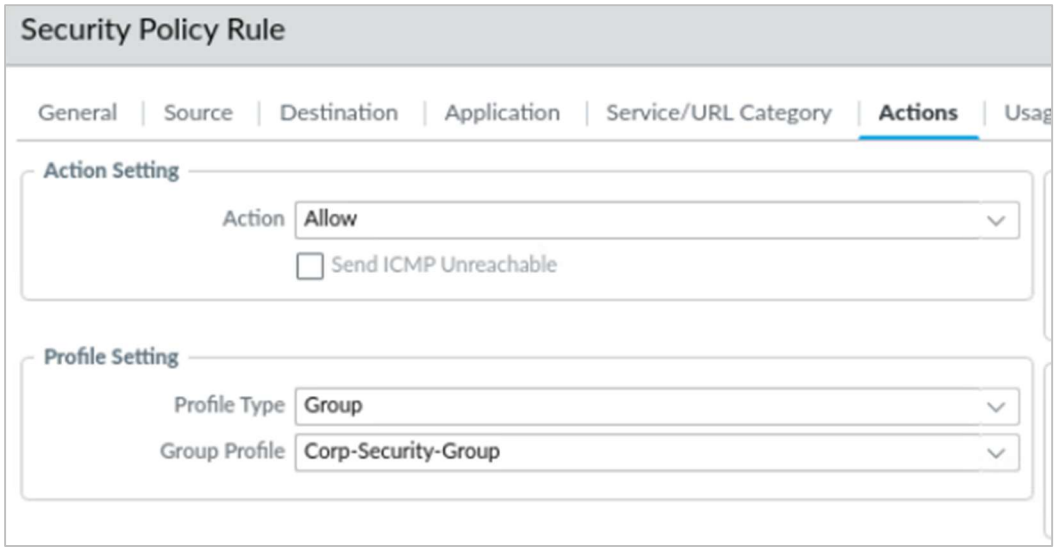


Users_to_Extranet, Users_to_Internet & Extranet_to_Internet all assigned to Corp-Security Group (3 Photos)

As the final step of the lab configuration, I opened the Actions tab for my three primary interzone policies—Users_to_Extranet, Users_to_Internet, and Extranet_to_Internet—and assigned the unified Corp-Security-Group. With this implementation complete, every permitted session crossing these zones triggers full deep packet inspection across all security engines simultaneously, successfully securing and locking down the enterprise perimeter architecture.







Conclusion & Engineering Summary

This capstone project successfully transformed a baseline virtual firewall template into a secure enterprise perimeter security architecture using a Palo Alto Networks firewall. Looking back at the work, I took a preconfigured setup and expanded it by setting up proper network segmentation, splitting traffic into separate zones for internal users, the outside internet, and an extranet partner segment. I configured virtual routing and source NAT so private internal IPs could safely reach the outside world without exposing the underlying network layout.

Next, I moved on to app id policies to control exact application traffic instead of just opening broad ports. I built out full threat inspection profiles covering antivirus, anti spyware, vulnerability protection, URL filtering, file blocking, and WildFire cloud analysis. Rather than assigning these individually to every single rule, I bundled them into a single profile group and applied it across all my main interzone policies so every piece of traffic gets thorough security inspection.

The best part of the workflow was running the verification tests to prove everything actually worked. I tested standard ping and DNS lookups, accessed secure web pages and internal servers, and confirmed that the firewall correctly blocked malicious sites like hacker9. Overall, this project gave me solid hands on practice with firewall administration, traffic routing, and threat prevention, making it a great baseline for my cybersecurity portfolio and a reliable reference for how everything fits together.